Empowering Medical Data Labeling for Non-Experts with DANNY: Enhancing Accuracy and Mitigating Over-Reliance on AI

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Abstract

Economic constraints on recruiting experts hinder efforts to build qualified datasets for utilizing AI in professional domains (e.g., medical diagnosis), which could provide societal benefits. To solve this issue, previous studies introduced crowdsourcing and AI to enable non-experts to perform expert-level data labeling. Yet, they encountered three challenges: 1) the limited applicability of crowdsourcing in less specialized domains (e.g., identifying animal species); 2) the chicken-and-egg problem, a paradox where high-performance AI is required to build a dataset to train such AI; and 3) over-reliance on AI, where non-experts, lacking expertise, may incorrectly label data when guided by sub-optimal AI. To address this, we introduce DANNY (Data ANnotation for Non-experts made easY), an AI-based tool designed to help non-experts label an arthritis dataset, aiming to increase labeling accuracy and mitigate over-reliance on AI. By externalizing a cognitive forcing intervention to foster critical thinking, DANNY provides two visualizations: 1) the Criteria phase, where non-experts define criteria across four arthritis features, and 2) the Correction phase, where they refine these criteria by comparing them to AI suggestions. In a study with 28 participants, DANNY users achieved higher accuracy and a more appropriate reliance on AI dependency than control groups. A follow-up study with 12 participants demonstrates how DANNY can be used to improve AI with an ensemble method. Our findings contribute new insights into using AI to support non-experts in labeling domain-specific data when expert resources are limited.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**.

Keywords

Non-expert, AI-enabled labeling tool, Dual-process theory

ACM Reference Format:
Youngseung Jeon, Christopher Hwang, and Xiang 'Anthony' Chen. 2025. Empowering Medical Data Labeling for Non-Experts with DANNY: Enhancing Accuracy and Mitigating Over-Reliance on AI. In *30th International Conference on Intelligent User Interfaces (IUI '25), March 24–27, 2025, Cagliari, Italy.* ACM, New York, NY, USA, 17 pages. https://doi.org/10.1145/3708359.3712161

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ACM ISBN 979-8-4007-1306-4/25/03
https://doi.org/10.1145/3708359.3712161

1 Introduction

The rapid advancement of artificial intelligence (AI) technologies has sparked interest across various fields, such as design [28, 30, 39], medical [23, 47], and business [29, 56]. Developing AI tailored to specific professional domains often requires a significant amount of annotated data [4, 40]. Ideally, engaging experts as annotators is crucial for reliable datasets, but recruiting them can be challenging due to their limited availability and high cost [35, 53].

In the HCI field, previous studies have proposed crowdsourcing systems that enable qualified labeling by engaging non-experts, who could be more easily recruited than experts. These works have helped non-experts improve labeling accuracy by identifying similarities and differences between classes in labeling, leading to reduced mislabeling [8–10, 19]. However, it remains questionable whether we can employ the crowdsourcing-based approach in highly specialized fields such as medicine [14, 60], because of the uncertainty and variance of non-experts' understanding of a professional domain they are unfamiliar with.

As a complementary solution, some researchers develop AI training methods that can learn from sub-optimal non-expert labeling in building datasets of professional domains. Examples include supporting non-experts in labeling disease-related objects, such as a tube on X-ray [15] and reducing noise from non-expert labeling during the AI training process [16, 62]. However, two problems remain. First, the "chicken-and-egg" problem: Using AI to help non-experts label data would require data in the first place to train such AI; therefore, AI models available to help non-experts are probably not yet well-trained (i.e., sub-optimal). Second, possible over-reliance on AI: Deploying AI introduces the risk of over-reliance on AI, since non-experts lack the knowledge and are likely to be misled by incorrect information from AI, leading to over-reliance [21, 45]. In other words, to practically assist non-experts in labeling data in professional domains using AI, it is necessary to utilize sub-optimal AI and provide an environment that enables non-experts to selectively extract appropriate information from AI for labeling tasks.

In this paper, we present *DANNY*, an AI-based annotation tool designed to help non-experts label arthritis in X-ray images within the medical domain. Specifically, we aim to utilize *sub-optimal* AI to assist non-experts in labeling and to design the system to *mitigate over-reliance* on AI. In other words, we investigate these two research questions (RQs):

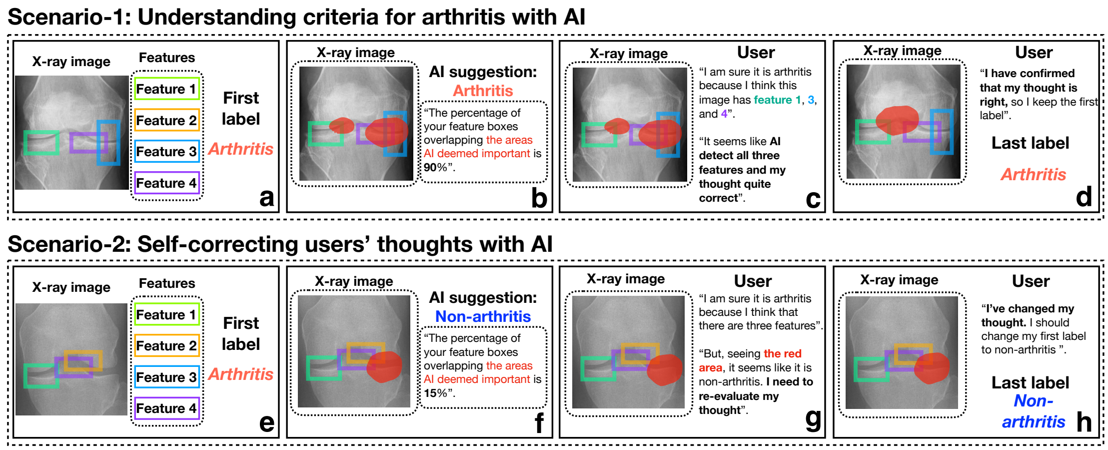

Figure 1: Two exemplar interaction scenarios (workflow diagrams, not the actual UI) of *DANNY* to support users' critical thinking. In Scenario 1, users identify the main key criteria for arthritis and make an initial label (a). They then review their understanding of the criteria with an AI suggestion, focusing on the red area that the AI highlights as important (b). Users compare their understanding of the criteria with AI's information (c). If users think their thoughts are reasonable, they maintain their original label (d). In Scenario 2, steps (e), (f), and (g) mirror steps (a), (b), and (c), respectively. However, if users believe that their understanding of the criteria is unreasonable after seeing AI's suggestions, they might critically reassess and self-correct their label (h).

- RQ1: How can we develop an AI-enabled tool that supports non-experts to label medical imaging data with increased accuracy AI while mitigating over-reliance on a sub-optimal AI?
- RQ2: How can such non-expert-labeled data be used to improve a medical AI model by providing additional training data to enhance its performance?

To answer RQ1, we applied the dual-process theory [32, 33] as a cognitive forcing intervention, which postulates that human thinking is governed by two distinct systems: a fast, automatic, and intuitive system (System 1) and a slow, deliberate, and analytical system (System 2). In *DANNY*, users first utilize System 2 (critical thinking) before using System 1 (heuristics) in labeling. When labeling arthritis on X-ray images, users are prompted to critically evaluate the presence of the four main criteria of arthritis (e.g., narrow joint space, osteophytes, irregularity of the cortical surface, and tibial spiking) [2, 49] before making a final labeling decision.

Building on the previous work by Buccinca et al. [6], we expanded the application of dual-process theory to real-world contexts by applying a cognitive forcing intervention. By externalizing two cognitive operations for critical thinking, we design *DANNY* to perform the intervention based on Lipman's theory [38] to support critical thinking: 1) thinking with criteria involving adherence to principles of logical validity, such as establishing a clear cause-and-effect relationship, and 2) self-corrective thinking, which includes objective thinking based on self-evaluation.

Our tool, *DANNY*, manifests these two conditions of critical thinking as non-experts' System 2 interactions with two visualizations—Criteria phase and Correction phase—when labeling arthritis on X-ray images. The Criteria phase enables users to check the presence of four criteria for arthritis [2, 49] and draw boxes in the area where they think that feature exists on an X-ray image, which prompts users to think with criteria before labeling. The Correction phase then allows users to verify and correct their boxes with AI suggestions (shown as highlighted important areas), which enables users to engage in self-corrective thinking. Figure 1 shows two exemplar scenarios of *DANNY*.

We conducted a user study to validate whether *DANNY* enables non-experts to accurately label arthritis datasets without over-relying on AI (Sec. 5). We recruited 28 participants, comprising an experimental group of 7 participants using *DANNY* and three control groups: 1) Ctrl-AI: relying solely on a sub-optimal AI, 2) Ctrl-XAI: using the same sub-optimal AI plus a saliency map indicating where AI deems important, and 3) Ctrl-no-AI: without any AI assistance. The reason to have both Ctrl-AI and Ctrl-XAI is to investigate explanation's impacts on over-reliance on AI.

Our findings observed a significant difference in the accuracy of labeling between the experimental and all the control groups. However, a comparison between Ctrl-no-AI and Ctrl-AI found no evidence of over-reliance on AI, although performance data do show that *DANNY* was able to help users develop *appropriate reliance* on AI, achieving significantly higher labeling accuracy than Ctrl-AI and Ctrl-XAI when AI was wrong.

To answer RQ2, we conducted the second user study to investigate how *DANNY* can be used to improve the accuracy of an AI model. Participants used *DANNY* to label arthritis on 600 X-ray images to increase AI's accuracy from a limited level (70%) to a reasonable level (80%) [44] (Sec. 6). Although relying solely on *DANNY* (with an improved AI accuracy of 74.62%) does not match the expert-generated ground truth (83.13%), an ensemble approach–where *DANNY* labels 300 low-uncertainty images and expert annotations were used for the other 300 high-uncertainty images–achieves an accuracy of 83.2%, approximating the all-labled-by-expert performance. This finding suggests that researchers who develop AI in a medical field can use *DANNY* to improve AI models while requiring fewer expert resources than the current approach.

This research's contributions include:

- We develop *DANNY*, the AI-based interface that assists non-experts in labeling medical datasets accurately while helping them appropriately rely on AI.
- We contribute a tool design that employs dual-process theory to solve the important machine-learning task of annotating a dataset by externalizing cognitive operations based on Lipman's theory.
- We demonstrate the applicability of *DANNY* in using non-expert-labeled data to improve an AI model with an uncertainty-based ensemble method.

2 Related work

2.1 Crowdsourcing-based Annotation Tool for Non-experts

Numerous HCI studies have suggested ways to enhance the quality of non-expert annotations through collaborative crowdsourcing. This collaborative method enables non-experts to identify patterns of similarity and difference in labeling by comparing their thinking to that of others, thereby enhancing the accuracy of dataset annotation. For example, Revolt [10] is a collaborative labeling approach that borrows from expert annotation workflows and involves a three-step process: voting, explaining differences in labels, and categorizing new concepts. Fang et al. [19] introduced a two-round crowdsourcing framework to enhance image labeling quality by having workers select multiple labels in the first round and choose the best one in the second round. Otani et al. [43] introduced a label aggregation method for hierarchical classification that organizes crowd workers based on their responses. Chang, Lee, and Igarashi [9] present Spatial Labeling, which leverages the spatial layout to improve label quality in non-expert image annotation. Chang et al. [8] present DualLabel, an annotation tool that enables secondary label assignments to images, enhancing annotation simplicity and model accuracy. However, this crowdsourcing-based approach faces significant challenges in expanding its applicability in real-world contexts. Variability in individual comprehension of specialized knowledge constrains its applicability to less specialized domains, such as the classification of specific animal species [8–10, 19]. Consequently, the effectiveness of non-experts' labeling becomes problematic in highly specialized domains, such as medicine.

In this work, we present *DANNY*, an AI-based annotation tool designed to help non-experts accurately label arthritis in X-ray images. To broaden the scope of a tool supporting non-expert labeling, we redefine the role of AI in labeling tasks by using sub-optimal AI as a tool that offers guidelines or feedback to assist non-experts in making their own decisions rather than providing a perfect answer. This approach enables non-experts to gain a deeper understanding of domain knowledge, such as the relationship between arthritis and its four main criteria, thereby enhancing the accuracy of their labeling.

2.2 Over-reliance on AI and Critical thinking

The AI community has presented AI frameworks to apply sub-optimal non-expert labeling in building datasets of professional domains: 1) supporting non-experts in labeling disease-related objects they are familiar with, such as a tube on X-ray [15] and 2) reducing noise from non-expert labeling during the training process on AI [16, 62]. In the first case, Damgaard et al. [15] extend annotation types by labeling the shortcuts to make the non-experts labeling meaningful. Instead of having non-experts directly label pneumothorax, they label the artifact, such as a tube, which is highly correlated with pneumothorax, thereby extending annotation types by labeling the shortcuts. In the second case, Zhang et al. [62] present a Tri-network learning framework to alleviate the problem of insufficient accurate annotations in a medical domain with non-expert annotations. The Tri-network learning framework consists of three networks where each pair of networks alternatively selects informative samples for the third network learning, according to the consensus and difference between their predictions. Amor, Silva-Rodríguez, and Naranjo[16] propose an uncertainty estimation-based framework to handle noise in the annotation process where non-experts label skin cancer. This is based on a novel formulation using a dual-branch min–max entropy calibration to penalize inexact labels during the training. However, there are two real-world constraints. First, using high-performance AI in the labeling process creates a paradox [22]: if the goal is to build a qualified dataset to develop high-performance AI, why label data when a high-performance AI already exists? This highlights the need to use sub-optimal AI when supporting an annotation task. Second, deploying sub-optimal AI introduces the risk of over-reliance on AI. Since non-experts lack professional knowledge, they are likely to be misled by incorrect information from the sub-optimal AI [21, 45].

In cognitive science, to mitigate over-reliance on AI, previous works presented cognitive forcing intervention that users should first engage System 2 (analytic thinking) before System 1 (heuristics) in the context of the dual-process theory [32, 33]. People become vulnerable to cognitive biases when they primarily rely on System 1 thinking, utilizing heuristics and shortcuts in decision-making [32]. On the other hand, System 2 (critical thinking) could be the way to reduce over-reliance on technology [20, 50, 51], while this is infrequently activated due to its slower pace and higher effort requirement. In human-computer interaction (HCI), Buçinca et al. [6] conceptually proved that cognitive forcing reduces over-reliance on an AI model. The authors used three types of cognitive forcing, such as on-demand, update, and wait, in general topics like cooking recipes.

Building on previous work, *DANNY* applies a cognitive forcing intervention to support non-experts in labeling within professional domains, such as medicine, by first engaging System 2 (critical thinking) before System 1 (heuristics). Based on Lipman's theory of critical thinking [38], we externalize two cognitive operations for critical thinking: 1) thinking with criteria, which involves adhering to principles of logical validity, such as establishing clear cause-and-effect relationships; and 2) self-corrective thinking, which includes objective evaluation and self-reflection. *DANNY* features two visualizations: 1) the Criteria phase, which enables users to check the presence of four key features of arthritis; and 2) the Correction

Table 1: Design goals for supporting critical thinking in the initial iteration.

Critical thinking	Requirement	Goal	Visualization	
			Initial version	*DANNY* (final)
Thinking with criteria	The thinking process of using standards to guide one's thinking	Provide an opportunity to **identify the cause-and-effect relationship between four main features and arthritis with AI.**	Reason phase	Criteria phase
Self-corrective thinking	The thinking process of evaluating and adjusting one's own thoughts and standards	Provide an opportunity to **reflect on one's own decision-making** and **encourage one to adjust if they look wrong**.	Comparison phase	Correction phase

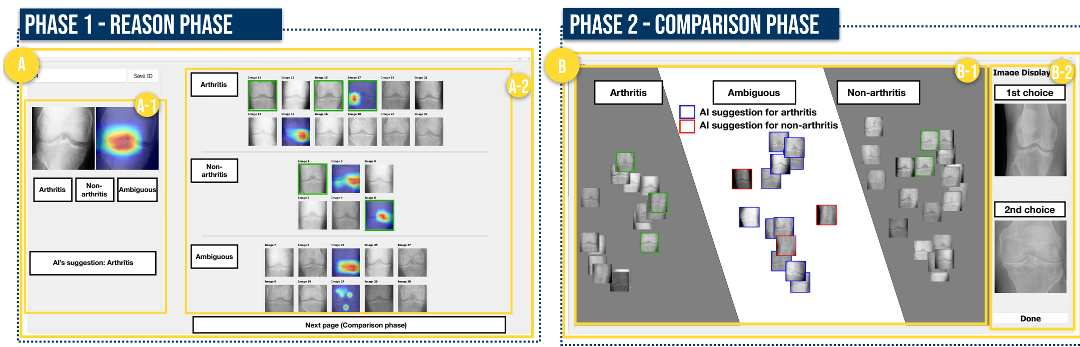

Figure 2: The initial version of the system. The initial version of the system presents two visualizations to support critical thinking: the Reason phase (A) and the Comparison phase (B). In the Reason phase (A), users confirm the relationship between the criteria and arthritis with AI, enabling them to understand the criteria. In the Comparison phase (B), users compare the characteristics of each group, allowing them to self-correct their understanding of the criteria.

phase, which allows users to verify and refine their thoughts using AI suggestions. This approach enables non-experts to engage in critical thinking, thereby improving an appropriate reliance on AI.

3 Iteration #1: Early Design of *DANNY* and Lessons Learned

We designed an initial version of the system to support critical thinking and improve accuracy while reducing over-reliance on AI in labeling arthritis X-ray images. Based on the two design goals in Table 1, the initial version consists of two visualizations: 1) the Reason phase and 2) the Comparison phase. In the pilot study, where six participants labeled arthritis with the initial version of sub-optimal AI (70%), we observed and summarized three lessons learned: L1) Drawing features is needed for understanding the criteria, L2) Showing AI intervention too early prevents self-corrective thinking, and L3) Batch processing protocol hinders self-correction.

3.1 The initial version of the system

Figure 2 shows the initial version of the system consisting of two visualizations: 1) Reason phase and 2) Comparison phase. We have designed two visualizations to satisfy two requirements for supporting critical thinking. The protocol for the initial versions of the system follows batch processing [46] where users first perform initial labeling of all 50 images in the Reasoning phase, then proceed to final labeling of them during the Comparison phase.

Table 2: The dataset used in the pilot and the first user studies is (a) the Total dataset (50 images). This dataset consists of two subsets — (b) The Subset-Correct (35 images) and (c) The Subset-Incorrect (15 images). (b) The Subset-Correct comprises images where the ground truth aligns with the AI suggestion. (c) the Subset-Incorrect dataset contains images whose ground truth does not match the AI suggestion.

Category	Dataset	# of images	Ratio (Arthritis: Non)
(a)	Total dataset	50	25 : 25
(b)	Subset-correct	35	17 : 18
(c)	Subset-incorrect	15	8 : 7

The specific protocol is as follows:

- Step 1: Users perform initial labeling on 50 X-ray images to determine the presence or absence of arthritis with AI suggestion and a saliency map indicating AI deems important in the suggestion in the Reason phase (Figure 2-A-1). If uncertain, users label the image as ambiguous. A labeled image is populated into Figure 2-A-2. Users label all 50 images.
- Step 2: In the meantime, users check the labeling results and understand the criteria by comparing images in each group (Figure 2-A-2). Additionally, users are supposed to choose a representative image for the four criteria in the arthritis

and non-arthritis groups (green boxes). The representative image refers to the images that clearly illustrate the criteria as understood by the users. Users can understand the criteria of arthritis through this step.

- Step 3: Users conduct the final labeling of the ambiguous-labeled images to arthritis-labeled and non-arthritis-labeled images in the Comparison phase (Figure 2-B). By comparing ambiguous images (red and blue boxes based on AI suggestions) to representative images (green boxes) and labeled images (Figure 2-B-1), users can self-correct their understanding of the criteria for arthritis. This step is performed after all 50 images have gone through steps 1 and 2.

The Reason phase (Figure 2-A) is a visualization that provides AI's suggestion of the presence of arthritis in an X-ray image with the saliency maps. Users can confirm the relationship between main features and arthritis with AI suggestions and reasons through saliency maps (Figure 2-A-1). We applied a saliency map because this is one of the intuitive methods that allow users to comprehend the results generated by AI systems in the medical domain. This map provides a rationale by directly presenting which part is the most important on the complicated medical images, such as X-ray and tumor, in the AI suggestion [5, 58]. In addition, by checking the common features of each group, they can understand the criteria for arthritis (Figure 2-A-2). We expected that this visualization would facilitate the first condition of critical thinking: thinking with criteria (Figure 3-E1).

The Comparison phase (Figure 2-B) is a spatial layout where users can efficiently compare images and then relabel images. Users can drag and organize the images and look at the images in close inspection (Figure 2-B-1) based on the advantage of a spatial layout [42, 48]. Unlike the Reason phase, the Comparison phase allows users to self-correct their understanding of criteria by checking the main features of arthritis they may have either misconceived or not previously considered. We expected that this visualization would support the second condition of critical thinking: self-corrective thinking (Figure 3-E2).

3.2 AI model and datasets

We built a dataset and trained models for the initial version of our system (OAI dataset [11]) consisting of the five Kellgren-Lawrence (KL) grades [34] from Grade 0 (Healthy) to Grade 4 (Severe arthritis). Since arthritis is defined as grade 2 or more on the KL grades [31], we defined Grade 0 (Healthy) as non-arthritis, and Grade 2 (Minimal), 3 (Moderate) and Grade 4 (Severe) as arthritis, removing the Garde 1 (Doubtful). The total number of images is 8,016 (arthritis: 4,159; non-arthritis: 3,857). After building the datasets for 70% for training (5,611), 10% for validation (802), and 20% for testing (1,603), we trained an AI model to predict arthritis with the dataset. Considering the potential applicability of our methodology, we endeavored to use a universally accessible model that anyone can utilize. In the prediction for arthritis, we used ResNet-18 [24] trained on the arthritis dataset. For the saliency map (XAI), we used Grad-Cam [54] that explains the predictions of any classifier to the classification process of ResNet-18 for the images.

Given that previous work states AI performance is considered reasonable between 80% and 95% [44], we utilize a model with 70% performance to simulate a real-world scenario, with the goal of developing AI with optimal performance (over 80%). To find the parameters that show limited performance (70%), we trained 63 models (ResNet-18) by incrementally and randomly adding 100 images to training and validation sets, starting from 100 images up to 6,300 images (the sum of train and validation sets) and using the test set (1,603 images). We identified that training with 600 images achieved the limited accuracy level of 70%, while training with 1,200 images resulted in the reasonable accuracy level of 83%. Overall, to demonstrate both optimal and sub-optimal performance, we used 2,803 images (1,200 for training and 1,603 for testing).

With the sub-optimal model, we built the dataset of 50 images for AI suggestions in this system (Table 2). Except for the 2,803 images used for proving optimal and sub-optimal level, we labeled the 5,213 images from the remaining training and validation sets first. Based on the requirements, such as the ratio of arthritis and non-arthritis, and the ratio of correct and incorrect suggestions in Table 2, we randomly selected 50 images. Furthermore, an additional 10 images, meeting the same requirements, were chosen for test sessions conducted prior to the labeling process (50 images), following the previous work on labeling assistance [17] that conducted a test session using about 20% of the total labeled images.

3.3 Pilot study

3.3.1 Participants. We conducted a pilot study with six participants to evaluate the initial version of the system. All participants were university students with no expertise in medicine (Table 4 in the appendix). To recruit participants from more diverse majors, we recruited people through online advertisements and snowball sampling. As compensation, participants received a 20 USD Amazon [1] gift card after the study.

3.3.2 Procedure. Before using the interface, participants were asked to watch the video to learn and understand the four features of arthritis (e.g., narrow joint space, osteophytes, the Irregularity of the cortical surface, and tibial spiking) [2, 49]. We made the video, referring to the X-ray materials from the radiology online course [2] to help them understand. After watching the educational video, participants conducted a test session where they labeled 10 images to prevent unintended errors due to unfamiliarity with the interface [17].

Next, participants were asked to label 50 X-ray images as either arthritis or non-arthritis in 30 minutes. Our user study aimed to determine whether participants using *DANNY* could label the arthritis dataset accurately while reducing over-reliance on AI. Table 2 shows (a) the Total dataset (50 images), which is composed of two datasets — (b) The Subset-Correct and (c) The Subset-Incorrect. (b) The Subset-Correct comprises images where the ground truth aligns with the AI suggestion. (c) The Subset-Incorrect contains images for which ground-truth does not match the AI suggestion. We hypothesize that our system effectively reduces over-reliance on AI and improves accuracy only if users can achieve high performance in both the entire dataset and the Subset-Incorrect. After participants completed the labeling tasks, we conducted an interview, during which the participants explained in detail whether/how AI

[1] https://www.amazon.com/
[2] https://www.radiologymasterclass.co.uk/

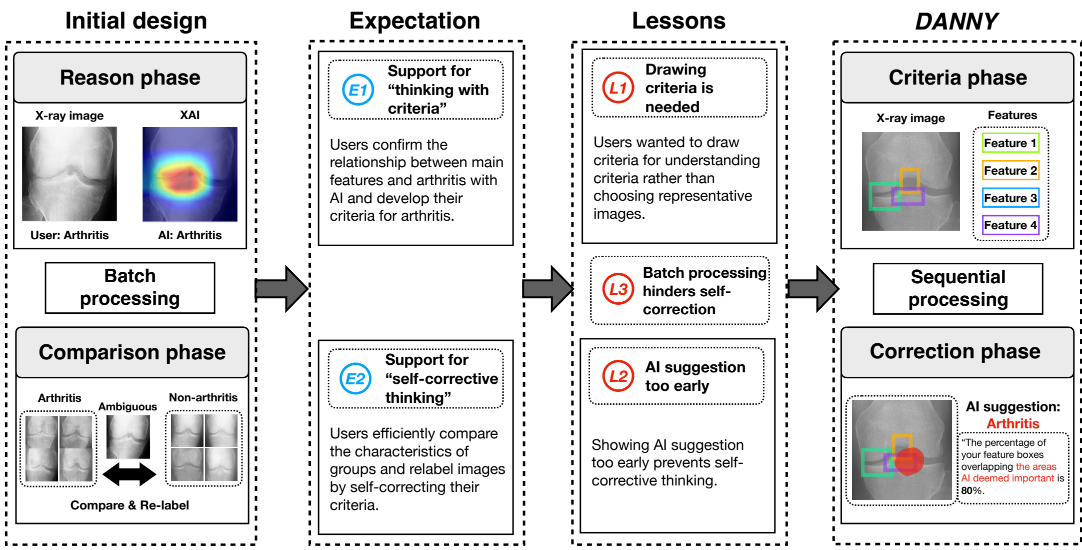

Figure 3: The iteration procedure from the initial design to *DANNY*. We built the initial design with the expectation (E1 and E2). After the pilot study, we gained three lessons (L1, L2, and L3). Addressing three lessons, we built *DANNY* consisting of two visualizations: 1) Criteria phase and 2) Correction phase.

helped them understand the criteria and correct them. When reporting interview quotes, we use P_p^X to denote participant number X in this pilot study (Table 4 in the appendix).

3.4 Results and Lessons

The average accuracy results are 68%, 79%, and 42% in the datasets (a), (b), and (c), respectively. This means that the initial version helped non-experts label at the suboptimal AI level (AI's performance: 70%). After the interviews were completed, we applied thematic analysis and iterative open coding [13] to analyze the interview transcripts. Two researchers coded and analyzed the transcripts for emerging themes, and the findings were discussed iteratively by the co-authors until a consensus was reached.

From the deployment experience and several pilots for design iterations, we developed several design lessons to help users better explore diverse opinion landscapes. In the following subsections, we introduce some of the lessons (denoted as L1-L3).

3.4.1 L1: Drawing criteria is needed for understanding criteria. In the initial version, participants were expected to understand the criteria for arthritis by choosing representative images during the Reason phase. Contrary to our expectations, participants wanted to annotate the x-ray image to highlight the criteria they found for understanding criteria for arthritis. For example: *"I think finding and recording the four main features would help me set my understanding of the criteria more. Also, it'd also be much more intuitive when I need to revise my understanding"* P_p^1). *"Choosing representative images just doesn't fully express my understanding criteria. I wish I could identify and record features."* P_p^2).

3.4.2 L2: Showing AI suggestion too early prevents self-corrective thinking. In the initial version, participants were supposed to understand the criteria for arthritis in the first visualization (Reason phase) and finally label ambiguous samples that were not labeled, self-correcting their understanding of the criteria in the second visualization (Comparison phase). However, contrary to our expectations, none of the participants used the second visualization as anticipated (relabeling only 1.5 images on average, 3% of the total 50 images). This indicates that the self-correction thinking was not supported as expected. Participants expressed a desire to first explore and identify the features on their own, before viewing the AI suggestion and the XAI image. For example: *"It feels like I leaned toward easily accepting what the AI suggested because I got exposed to its information before I could think things through on my own"* P_p^6). *"I think providing AI suggestions with saliency maps was quite reasonable. I completed labeling 49 images in the first visualization"* P_p^5). In addition, previous works support these comments that providing AI results before humans make a decision leads to over-reliance on AI [20, 50, 51].

3.4.3 L3: Batch processing protocol hinders self-correction. In the initial version, participants were supposed to use the second visualization after finalizing labeling 50 images in the first and reviewing them (batch processing protocol). We anticipated that users would understand the criteria by selecting representative images from 50 images before self-correcting their understanding of the criteria in the second visualization. However, users expressed their difficulty using the second visualization because they could not recall why they could not label those images in the first visualization. For example, *"I really struggled to go back and tweak my understanding of the criteria in the second feature. I just couldn't recall why labeling* *was tough in the first function, and without knowing that, I ended up not changing much at all in the second round."* P_p^4).

4 Iteration #2: Final Design of *DANNY*

Figure 3 shows the overview of the iteration procedure from the initial design to *DANNY*. Reflecting on the three lessons (e.g., L1, L2, and L3), we iterated on the design of *DANNY* to provide two new visualizations – Criteria phase and Correction phase. To address L1, we revised the UI so that users can draw a box to represent their thoughts regarding four criteria for arthritis to understand the criteria (L1). For L2, we defer AI suggestion to the second visualization instead of the first visualization for self-correction. For L3, we changed the protocol from batch processing to sequential processing, in which users complete initial and final labeling one image at a time.

4.1 Criteria phase (L1 and L3)

The Criteria phase allows users to establish the cause-and-effect relationship between four criteria (e.g., narrow joint space, osteophytes, the irregularity of the cortical surface, and tibial spiking) [2, 49] and arthritis to support thinking with criteria, addressing L1, which states that drawing criteria is needed to understand criteria for arthritis. In addition, to address L3, we applied sequential processing, where users initially label an image (Criteria phase), then perform the final labeling of the image (Correction phase), and afterward move to the next image.

The use of the Criteria phase follows the steps outlined below:

- Step-1-1: Users check the presence of criteria. When finding a criterion, they click a drop-down to choose *present.* If the criterion is absent, they click the drop-down and select *not present.* If the user is unclear about the feature's existence, they can select *ambiguous* (Figure 4-A).
- Step-1-2: Users draw a box where they think a feature exists by dragging it with the left mouse button. If users draw a box incorrectly, they can delete it by hovering the mouse over it and right-clicking (Figure 4-B).
- Step-1-3: Step-1 and Step-2 are repeated until users check all four criteria, after which they can proceed to the next step.
- Step-1-4: Labeling an X-ray image for the initial labeling (Figure 4-C) based on the criteria specified in the previous steps.

Through these steps, we expect that users will have opportunities to learn the cause-and-effect relationship between criteria and arthritis to support their thinking with criteria (System 2).

4.2 Correction phase (L2)

To address L2, the Correction phase offers an AI suggestion for labeling after finding the criteria. This visualization provides three types of information to support self-corrective thinking: 1) an AI suggestion for labeling (Figure 4-D), 2) the red area that AI deems important for the suggestion (Figure 4-E), and 3) the overlap percentage between the boxes the user made and the red area (Figure 4-F). If the information provided by the AI is reasonable, it is accepted for final labeling; otherwise, the original labeling is retained, following these steps.

- Step-2-1: Users check AI suggestions for labeling arthritis or non-arthritis (Figure 4-D). Based on overlapped percentages between the AI box and boxes drawn by users, if the IoU is 50% or less, this says "The AI seems to consider different parts as important"; if the IoU is over 50%, this says "The AI seems to consider the part you think art important as important."
- Step-2-2: Users compare the boxes they drew in the Criteria phase to the red area that AI deems important in AI's labeling suggestion (Figure 4-E). This comparison enables users to evaluate the appropriateness of AI suggestions by comparing the red area by AI with the criteria that users found.
- Step-2-3: Users check the overlapped percentages between each box they drew and the red area from AI. This information helps to understand at a glance how much the AI criteria and personal criteria quantitatively align (Figure 4-F). The overlapped percentage is calculated by comparing the number of red pixels scanned from the AI's saliency map within the area of the drawn boxes and dividing that value by the total number of red pixels of the AI's red focus area.
- Step-2-4: Users proceed with the final labeling of an image by reviewing their understanding of the criteria. If users believe their understanding of the criteria is reasonable, they maintain their original label. If not, they critically reassess and self-correct, ultimately changing their label (Figure 4-G).

We expect that users will have opportunities to adjust their understanding of the criteria for arthritis based on AI suggestions and feedback, thereby supporting their self-corrective thinking.

5 RQ1: Can *DANNY* Increase Labeling Accuracy & Mitigate AI Overreliance?

We conducted two user studies, each to answer two research questions. For answering RQ1, our first user study involving 28 participants aimed to determine whether participants using *DANNY* could label the arthritis dataset with high accuracy while reducing over-reliance on AI.

5.1 Participants

We recruited 28 non-expert participants into one experimental group of 7 and three control groups, each comprising 7 participants. All participants were non-experts in the medical field and were students from the same university. Table 5 in the appendix shows the detailed demographic information of participants. To recruit participants from more diverse majors, we recruited people through online advertisements and snowball sampling. Our study was approved by the Institutional Review Board (IRB), and the consent of the participants was obtained before the study. As compensation, participants received a 20 USD Amazon [3] gift card after the study.

5.2 Design & procedure

The study procedure is the same as in the pilot user study (Sec. 3.3.2). After watching the educational video on the criteria for arthritis, participants labeled 10 images as a test. Then, they label 50 images (Table 2) in 30 minutes.

We conducted a between-subjects study with one experimental and three control groups. In the experimental group, non-expert participants were asked to use *DANNY* to label the 50 arthritis

[3] https://www.amazon.com/

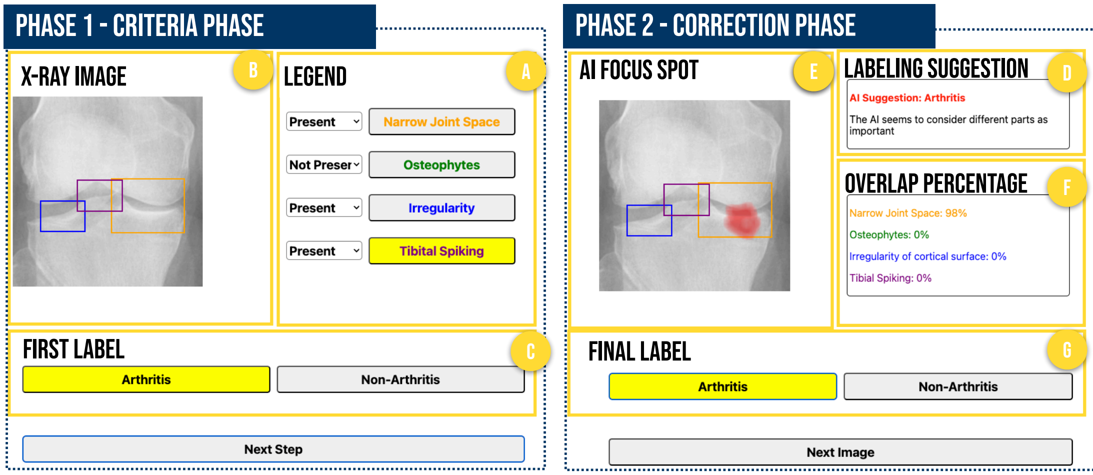

Figure 4: *DANNY* presents two visualizations to support critical thinking: 1) the Criteria phase and 2) the Correction phase, addressing three lessons from the pilot study with the initial version.

Figure 5: The interfaces were designed for three control groups: (a) a control group without AI suggestions (Ctrl-no-AI), (b) a control group with AI suggestions (Ctrl-AI), and (c) a control group with AI suggestions and saliency maps indicating areas the AI deems important (Ctrl-XAI). In both Ctrl-AI and Ctrl-XAI, participants can view the AI's suggestions, which provide information about the presence of arthritis in the images. In contrast, participants in the Ctrl-no-AI group cannot access any AI suggestions during the task.

X-ray images. In the first control group, non-expert participants labeled the dataset without any AI suggestions (Ctrl-no-AI). In the second control group, non-expert participants labeled the dataset with AI suggestions (Ctrl-AI). In the third control group, non-expert participants label the dataset with suggestions from AI and saliency maps indicating AI deems important in the suggestions (Ctrl-XAI).

The rationale behind the three control groups is as follows: Ctrl-no-AI serves as a baseline, while Ctrl-AI and Ctrl-XAI are designed to examine the effect of explainable information, such as a saliency map, on mitigating over-reliance on AI. Explanations can help users better evaluate the accuracy of AI recommendations and understand how AI systems function [18, 26, 52], while detailed explanations can sometimes cause users to develop inappropriate reliance [61]. Figure 5 shows the dummy interfaces for control groups.

5.3 Measurement

To answer RQ1, we examined the impacts of *DANNY* on increasing accuracy while mitigating over-reliance on AI. In subsection 5.4.1 and 5.4.2, we analyze the accuracy of (a) the Total dataset and (c) the Subset-Incorrect in Table 2, respectively, across the experimental and control groups.

5.4 Results

5.4.1 Increased accuracy. To demonstrate that *DANNY* improves labeling accuracy, we evaluate whether there are significant differences in labeling accuracy between the experimental and control groups. A one-way ANOVA was conducted to measure whether *DANNY* helped increase accuracy. Post-hoc comparisons were conducted using Tukey's correction [57]. Figure 6 shows that there is a significant difference in the accuracy between the experimental and all control groups (Ctrl-no-AI, Ctrl-AI, and Ctrl-XAI) in the dataset.

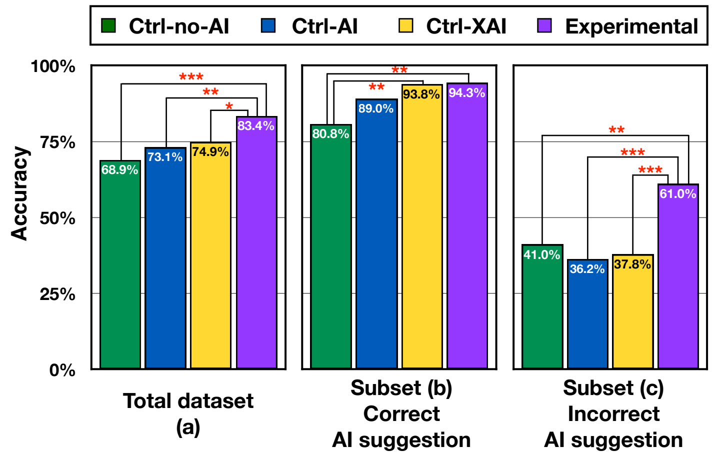

Figure 6: Bar plots showing the accuracy differences between the experimental and control groups in the dataset (Table 2) ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$).

In the case of the total dataset (50 images, Figure 6-(a)), the accuracy between the experimental group and control groups showed significantly different accuracy ($F(3,24) = 10.4$, $p <. 001$). According to the post-hoc analysis, there are significant differences between the experimental and all control groups: 1) Ctrl-no-AI ($p < .001$), 2) Ctrl-AI ($p = .004$), and 3) Ctrl-XAI ($p = .019$). There was no significant difference between the control groups. These results verify the hypothesis that *DANNY* impacts increasing accuracy.

In the case that AI suggestions are correct (35 images, Figure 6-(a)), the accuracy between the experimental group and control groups showed significantly different patterns ($F(3,24) = 7.16$, $p = .001$). According to the post-hoc analysis, the experimental and the Ctrl-no-AI groups showed significant differences ($p = .002$). The Ctrl-XAI and the Ctrl-no-AI groups showed significant differences ($p = .003$). Given that both the experimental and Ctrl-XAI groups provide suggestions from AI and saliency maps highlighting important factors for the suggestions, these elements can enhance accuracy when the AI's suggestions are correct.

The difference in usage time between the experimental group and the three control groups may indicate the potential of *DANNY* for improving accuracy. The average usage time per image of *DANNY* is 40.04 seconds, which was significantly longer compared to the Ctrl-no-AI group (5.16 seconds), the Ctrl-AI group (6.45 seconds), and the Ctrl-XAI group (5.7 seconds). These results suggest that non-experts using *DANNY* spent sufficient time identifying the labeling criteria, leading to a more deliberate labeling process and higher accuracy.

5.4.2 DANNY enabled appropriate reliance on AI. Over-reliance on AI is defined as users accepting incorrect AI recommendations [45]. With the definition, to demonstrate the effect of *DANNY* on mitigating over-reliance on AI, we evaluate the Subset-Incorrect (15 images, Figure 6-(c)). According to the post-hoc analysis, there are significant differences between the experimental and all control groups: 1) Ctrl-no-AI ($p = .003$), 2) Ctrl-AI ($p < .001$), and 3) Ctrl-XAI ($p < .001$). There was no significant difference between the control groups.

The accuracy of the Ctrl-no-AI group is not significantly higher than that of the Ctrl-AI or Ctrl-XAI groups, indicating that participants' performance did not significantly decline when using sub-optimal AI compared to working without AI. Therefore, this result challenges our assumption that non-expert users would over-rely on a sub-optimal AI. On the other hand, there is a significant difference between the experimental group and two control groups with AI suggestions (Ctrl-AI and Ctrl-XAI) in the Subset-Incorrect dataset, suggesting that *DANNY* was able to help users develop *appropriate reliance* on AI, i.e., users can correctly override wrong AI suggestions to reach the correct labels [45, 52].

5.5 Interview

We conducted interviews with the seven participants in the experimental group. During the interview, the participants explained in detail how AI helps users label arthritis X-ray images, helping them understand the criteria for arthritis (*the Criteria phase*) and correct their understanding of the criteria by themselves (*the Correction phase*). When reporting interview quotes, we use P_u^X to denote participant number X in the user study.

5.5.1 Supporting critical thinking for non-experts. All participants answered that the *DANNY* provided sufficient support for critical thinking in the labeling process. They mentioned that the two visualizations enabled them to have enough time to formulate their understanding of the criteria for arthritis and opportunities for self-correction. Regarding *the Criteria phase*, participants mentioned that this visualization was useful in understanding the relationship between arthritis and the four criteria. For example, *"I was able to understand criteria by thoroughly exploring the features in the Criteria phase. Since I am not a medical professional and only learned from the short introductory video, by the Criteria phase, I could more understand what arthritis is"* (P_u^3). *"Boxing the criteria directly into boxes really helped me understand them better. If I had* *just looked at them, I think it would have been hard to remember the criteria. Especially with osteophytes, which was hard to grasp, but intentionally boxing them over and over gave me a bit more confidence."* (P_u^1). *"Boxing the criteria was helpful in understanding the criteria for arthritis. In the next phase, this understanding led me to consider whether specific criteria exist when evaluating AI suggestions. Without this boxing of the criteria, I might have relied on my intuition alone to judge the AI's suggestions"* (P_u^1). Regarding *the Correction phase*, participants mentioned that this visualization was helpful in self-correcting their criteria for arthritis. For example, *"Looking at the AI's suggestions and the red areas it considered important helped me adjust my understanding of the criteria, which overall improved my understanding of arthritis"* (P_u^2). *"Comparing it with the red area helped me understand the AI's suggestions. Like, when it suggested 'non-arthritis' even though it clearly looked like arthritis, the saliency map showed the AI didn't really focus on the criteria. After going through this a few times, I feel like I got better at filtering the AI's suggestions"* (P_u^7).

5.5.2 Impact of User Confidence on AI Suggestion Acceptance.

Participants mentioned that the level of their confidence in the Criteria phase impacts the acceptance of AI suggestions in the Correction phase. In the case of high confidence, users could evaluate the AI suggestions and important areas based on their criteria. For example, *"If I was confident in my initial labeling, I could assess the suitability of the AI's red box for final labeling"* (P_u^1). *"When I was confident in the initial labeling and the red box seemed odd, I didn't consider the AI suggestion and proceeded with final labeling."* (P_u^4). On the other hand, when users have low confidence in their first labeling, they mention that the impact of AI suggestions is limited. For example, *"Considering AI performance is about 70% when I was not sure of my initial labelings, it was hard to understand AI suggestions"* (P_u^2). *"When receiving assistance from AI without understanding the presence of features, it seems I made heuristic rather than critical decisions."* (P_u^7).

5.5.3 Features AI detected.

Participants demonstrated interest in AI's capabilities regarding reasoning, such as which criteria AI detects for the suggestions. Users noted that showing the features AI detected would be more beneficial in comparing their criteria to AI suggestions which is important for self-correction. For example, *"Because the AI isn't all that great, knowing which features it's picking up could really help understand AI's suggestion."* (P_u^6). *"The red area is really helpful, but in the end, it seems crucial to know which features the AI is recognizing. This would be incredibly beneficial in adjusting my own criteria."* (P_u^1).

6 RQ2: Can data labeled with *DANNY* be used to improve AI?

To answer RQ2, we conducted the second user study with 12 participants to investigate whether non-expert labeled data with *DANNY* can contribute to improving AI performance during the development process to a reasonable level. We devised a scenario in which non-experts use *DANNY* to label arthritis dataset to improve AI with sub-optimal performance (70%) towards a level above 80% (Sec. 6.1). We evaluated the impact by comparing the accuracy of models trained with two datasets labeled by: 1) non-experts using *DANNY* and 2) ground-truth (previously obtained from experts). To improve the applicability of *DANNY*, we presented an ensemble labeling approach based on uncertainty: high-uncertainty images are labeled with ground-truth from experts, while low-uncertainty images are labeled by non-experts using *DANNY* (Sec. 6.4). This result indicates that our ensemble method, *DANNY*, can enhance AI performance while maintaining satisfactory results and potentially reducing the number of experts required.

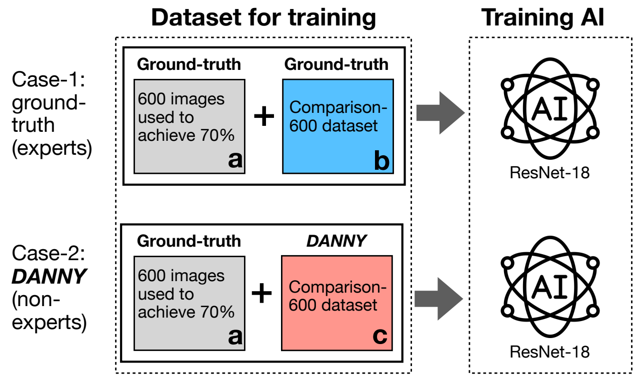

Figure 7: The setup of the second user study in the scenario where the goal is to increase AI performance from a limited level (70%) to a reasonable level (over 80%) with *DANNY*. We created two versions of the Comparison-600 dataset: b) ground truth (previously labeled by experts) and c) labeled by *DANNY*. We then merged each with a) the 70% dataset, respectively, to compare the outcomes of AI performance.

Table 3: The Comparison-600 dataset used in the second user study. All ratios regarding information align with Table 2. (b) The Subset-Correct dataset comprises images in which the ground-truth aligns with the AI suggestion. (c) the Subset-Incorrect dataset contains images whose ground truth does not match the AI suggestion.

Category	Dataset	# of images	Ratio (Arthritis: Non)
(a)	Total dataset	600	300 : 300
(b)	Subset-Correct	420	210 : 210
(c)	Subset-Incorrect	180	90 : 90

6.1 Setup

In our experimental scenario, the goal is to increase AI performance from limited performance (70%) to a reasonable level (over 80%) using *DANNY*. Figure 7 shows the setup for the second user study to implement the scenario. To achieve this, we first need to determine the range of the number of datasets needed for increasing the performance from 70% to 80%. In Section 3.2, we experimentally identified the range from 600 to 1,200 images. In other words, we would first train an initial AI model with 600 images to achieve

70% accuracy and then add another 600 images (some of which were labeled by non-experts using *DANNY*) to improve that AI. Then to another 600 images, we built the Comparison-600 dataset. We randomly chose 600 images satisfying the ratio requirements, including the ratio of arthritis and non-arthritis, and the ratio of correct and incorrect suggestion, as shown in Table 3. We created two versions of labels for the Comparison-600 dataset: 1) by ground truth (previously obtained from experts) and 2) non-experts using *DANNY*. Then, each version was merged with the 70% dataset to compare the AI outcomes (ResNet-18 [24]).

6.2 Participants and procedure

We recruited 12 participants who were all non-experts in the medical field (Table 5 in the appendix). In the recruitment, we used the same method as the first user study (Sec. 5.1) but none of the participants were in the first study. Except for the dataset, the procedure for the second user study was identical to that of the pilot (Sec. 3.3.2) and the first user study (Sec. 5.2).

6.3 Results: improving AI only using non-expert-labeled data

In this section, we report two results: 1) the labeling accuracy of the Comparison-600 dataset by twelve participants, and 2) the performance of a model trained with the Comparison-600 dataset labeled by non-experts with *DANNY* and ground-truth. First, the labeling accuracy of the Comparison-600 dataset by twelve participants is 76.5%. This is higher than the performance of AI in *DANNY* (70%) while lower than the first user study (83.4%). To understand the performance difference between the two studies, it should be noted that in the first study, all participants labeled the same 50 images, while in the second study, each of the 12 participants labeled a different batch of 50 images, resulting in a total of 600 images. The number of images in the second user study is 12 times greater than in the first. This increase includes more ambiguous images to label, which may have influenced the accuracy of the second study.

Next, with the newly-labeled dataset, we trained AI models and evaluated them. The accuracy of model training with the Comparison-600 dataset labeled by non-experts using *DANNY* is 74.62%. The ground-truth version (83.13%) is higher, indicating that using non-expert-*only* labels might be challenging in improving AI models. However, our ensemble approach in the next section, which involves applying non-expert labeling to images with low uncertainty and using ground-truth labeling for the remaining images, demonstrates the potential benefits of non-expert labeling using *DANNY*.

6.4 Results: improving AI using an uncertainty-based ensemble labeling approach

To identify a possible direction in which non-experts who lack specialized knowledge can contribute to the labeling process, we evaluated accuracy by the uncertainty score of the datasets, based on the concept that the uncertainty of labeling an image indicates complexity and difficulty of the data in the labeling process [25]. Our assumption is that if non-experts label images with low uncertainty, they can label them more accurately due to the relatively lower complexity following the results. We confirmed the relationship between labeling accuracy and uncertainty based on the results from the first user study (Sec. B.1 in the appendix).

Building on these results in the first user study, we created three scenarios based on uncertainty in the second user study: 1) the low-high (blue), 2) the high-low (green), and 3) the unordered (grey). In the low-high scenario, non-expert-labeled images were used, starting with those having the lowest uncertainty and following an uncertainty-ascending order. For instance, at the x-axis point of 50 in Figure 7, the model was trained using the 50 images with the lowest uncertainty labeled by non-experts, while the remaining 550 images were labeled with ground-truth labels. Conversely, in the high-low scenario, non-expert-labeled images were applied in the opposite order, starting with the highest uncertainty, following an uncertainty-descending order. In the unordered scenario, non-expert labeling was applied randomly without considering uncertainty.

Figure 8 shows the accuracy of an AI model as it is trained with increasing numbers of images labeled by non-experts, out of a total of 600 images in the three scenarios. Overall, all three scenarios show a trend where accuracy decreases as the number of images labeled by non-experts increases. However, the Lowest uncertainty scenario (Figure 8-blue line) shows the applicability of non-expert labeling. As an example, when non-experts labeled the 300 images with the lowest uncertainty and ground-truth was applied to the remaining 300 images, the performance matched the figure of 83.13% (the dashed red line in Figure 8), which was achieved by training entirely with ground-truth labeled data. This finding indicates the possible application of *DANNY* in improving AI models lies in reducing the need of expert participation using the ensemble approach: experts focus on the high-uncertainty images while non-experts cover the middle- and low-uncertainty images with *DANNY*.

7 Discussion

We have outlined our research methodology and introduced *DANNY* by incorporating the critical thinking theory for the applicability of an AI-based system to increase labeling performance while mitigating over-reliance on AI (RQ1), and then we proved the impact of *DANNY* through the results in the first user study. Through the second user study, we demonstrated the feasibility of *DANNY* in developing an AI model (RQ2). We summarize the findings of the study and discuss their implications. We also report the study's limitations and our plans for future research.

Externalization of critical thinking theory in *DANNY*. In this research, we first identified two primary design rationales to facilitate labeling from the Lipman's critical thinking theory [38]: (1) thinking with criteria and 2) self-corrective thinking. In theory, understanding and developing criteria are crucial for enhancing critical thinking. By externalizing this theory, we developed *DANNY* that employed a cognitive intervention to enable users to understand the criteria and self-correct them. Understanding and developing the criteria equips users with the interpretive ability to consume AI suggestions properly. For example, in our user study,

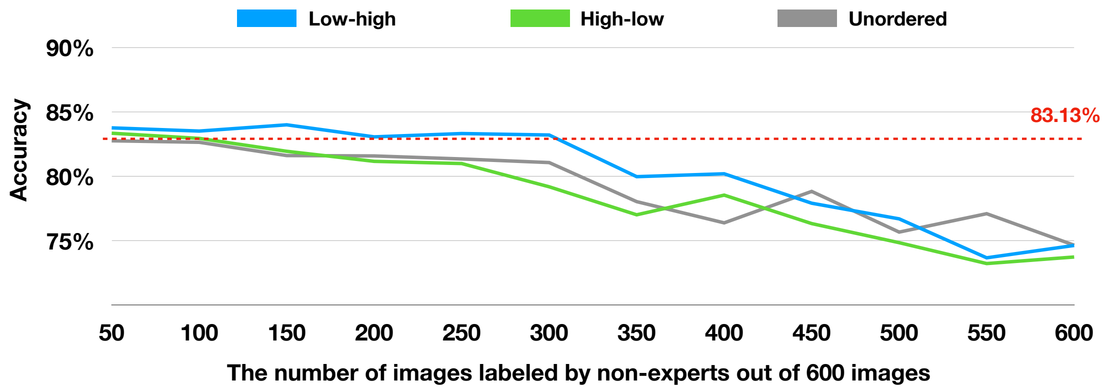

Figure 8: The graph illustrates the accuracy of an AI model trained with increasing numbers of images labeled by non-experts, from a total of 600 images across three scenarios: 1) the low-high (blue line) where images are ordered by increasing uncertainty; 2) the high-low (green line) where images are ordered by decreasing uncertainty; and 3) the unordered (grey line) where the order is random. The red dashed line represents the AI's performance, achieving an accuracy of 83.13% when trained on all 600 images labeled as ground truth. Notably, the low-high scenario shows performance comparable to the ground-truth level when using up to 300 images, achieved with the most non-expert labeling (300 images) among the three cases.

boxing the criteria in *the Criteria phase* not only helped users understand the criteria for arthritis but led them to utilize criteria when evaluating AI suggestions. Through labeling 50 images, they iterated and deepened their understanding of these criteria. Users then assessed whether the AI suggestions are correct or incorrect based on their understanding of these criteria, which led to appropriate reliance on AI. Conversely, when users lack a solid understanding of either the criteria or arthritis labeling, interacting with AI might lead to inappropriate reliance on the AI. This suggests that critical (or System 2) thinking by means of thinking in criteria matters in their interaction with AI.

Enabling appropriate reliance on AI and domain knowledge. Over-reliance on AI occurs when users accept incorrect AI recommendations, leading to an overall performance no better than not having AI. Even though we found no statistically significant evidence of over-reliance on AI, we did show that *DANNY* can help non-experts develop appropriate reliance on AI. With *DANNY*, users overrode wrong AI suggestions (self-reliance) and followed correct AI suggestions (AI-reliance), leading to overall better decision-making for labeling than all three baselines. In essence, the *DANNY*'s ability to enable appropriate reliance stems from its capacity to help non-experts to understand domain knowledge, such as the four criteria for arthritis. Previous works' findings align with this point. For example, Wang and Yin [59] emphasized the importance of designs that enhance understanding of the model, its inherent uncertainty, and trust calibration to improve AI-human interaction in decision-making processes. While these three properties have been demonstrated in domains requiring relatively less expertise (e.g., predicting criminal recidivism), their effectiveness remains limited in specialized domains that require domain knowledge (e.g., forest cover prediction). Furthermore, Cai et al. [7] have shown that providing experts with opportunities to refine AI predictions—such as emphasizing critical parts of the model's output based on their own criteria—can influence trust in the algorithm and decision-making. In specialized fields such as medicine, this suggests that when supporting human decision-making, the priority lies in supporting domain-specific knowledge. *DANNY* to enable understanding criteria for decision-making in a professional field suggests its potential to help prevent or mitigate over-reliance, thereby preventing or mitigating over-reliance and improving decision-making.

***DANNY*'s generalizability to other domains.** In this work, we focused on arthritis X-ray images as a case study of *DANNY*'s design based on dual-process and critical thinking theory. We expect the workflow supported by *DANNY* is extensible to other labeling tasks, from other medical images, to a range of scientific data where the labelers need to employ critical thinking of key criteria.

Specifically, to implement *DANNY*-like workflow for labeling data in other professional domains, future work needs first consider the following questions:

- Whether the number of criteria is manageable for non-experts to understand.
- Whether the standards between criteria are clear enough for non-experts to comprehend.

If these conditions are met, *DANNY* can be applied to other domains following four steps. First, researchers need to define the cause-effect relationship in the task. For example, when developing an AI-based interface to find tumors, researchers should define the tumor criteria, such as the number of nuclei and necrosis probability [23]. Secondly, for building *DANNY*, researchers should develop AI models having sub-optimal performance (60-70%). Thirdly, researchers should provide enough chances and time for non-experts to understand the criteria before AI supports as the medium for interaction with AI. Lastly, AI should provide labeling suggestions and the causes to support self-correction for their understanding of criteria.

The usage of *DANNY* in developing AI. We proposed the ensemble approach based on uncertainty grouping through the second user study, whereby *DANNY* can support non-experts to contribute up to one-third of the total dataset. In our second study, the difference between building the full expert-labeled dataset and including non-experts' contribution was negligible (about 0.8%). This approach strategically utilizes AI uncertainty to allocate labeling tasks, aligning with the principles of Active Learning [1, 55], which aim to maximize the use of limited labeling resources to improve model performance. This approach can be directly integrated into an Active Learning framework that can optimize resource allocation based on uncertainty, achieve cost efficiency, and improve model performance through a streamlined workflow. In addition, applying human-in-the-loop (HITL) [41] shows the potential for increasing non-experts' contribution, given that they can interact with AI that has a better performance by expert labeling. When experts label high-uncertainty images and these data are used for AI training, the resulting AI achieves higher performance, enabling more effective collaboration with non-experts during labeling tasks. Considering the volume of data emerging across various domains, such as tumors [23], genomics [37], and satellite data [27] and the corresponding increase in the demand for experts, such an ensemble approach presents the potential to save time and money of building datasets in the real world, maintaining the quality of labeling.

Intuitive comparison for self-correction. Non-experts should continually refine and develop an understanding of criteria with clear explanations for AI's suggestions, assisting them in reducing over-reliance on AI while enhancing the accuracy of their labeling tasks. During interviews, most participants preferred knowing the detected features by AI for its suggestions to important areas (e.g., red areas). Knowing detected features enables final labeling through intuitive comparison with their identified features. In essence, they desired information similar to their understanding of criteria when receiving AI suggestions. Previous studies have addressed local feature importance within AI interfaces that support human decision-making [36]. Local explanation methods clarify the reasoning behind particular predictions, helping individuals comprehend these predictions to make well-informed decisions. Our research emphasizes the importance of aligning the level of information provided by AI with the underlying reasons for users' criteria, supporting self-correction, a critical aspect of critical thinking.

Limitations and future work We acknowledge the limitations of our study, which we plan to address in future work. First, the reason for the AI labeling suggestion may have limitations in supporting non-experts' needs. *DANNY* only shows the area why AI thinks it is important for a particular labeling suggestion but does not further specify the detected features users wanted for intuitive comparison with their understanding of the criteria. To address this, we plan to expand *DANNY* with AI models predicting labeling and visualizing criteria. Beyond simply highlighting specific parts that AI deems most important, the future system can show visual attributes by which AI identifies the criteria. This allows users to interact with the AI based on that criteria prediction information, facilitating a deeper understanding of AI's decision-making process and potentially leading to a more appropriate reliance on AI. Gu et al. [23] presented an AI-based interface to help a pathologist find a tumor by developing multiple AI to detect the criteria for tumor, such as the number of nuclei and necrosis probability. In this study, the absence of a dataset related to criteria limited the detection of criteria. However, in future research, through expansion studies in domains where data can be secured, such as tumors, we plan to broaden the potential applicability of *DANNY*.

Secondly, although 46 people participated in the pilot and two user studies, these results may only represent a specific demographic. Participants in our study were either currently enrolled in a university or had graduated from one. Given the importance of learning abilities in our research, we must consider the influence of particular demographics, such as educational levels. In future research, we aim to recruit users from more diverse backgrounds to test the applicability and limitations of *DANNY*. Additionally, to reduce errors inevitably arising from individual differences, we plan to develop an interface that allows multiple non-experts can collaboratively label a single image.

8 Conclusion

Our study proposed a new methodology to empower non-expert labeling of data in a professional domain with AI suggestions. Through the implementation of a cognitive forcing intervention and critical thinking workflow, *DANNY* helps non-experts accurately label arthritis in X-ray images, while facilitating appropriate reliance on AI. To achieve this, *DANNY* provides two visualizations: 1) the Criteria phase, which enables users to build their understanding of the criteria for arthritis, and 2) the Correction phase, which allows them to self-correct their understanding of the criteria. Both the quantitative and qualitative results of our user studies confirmed that *DANNY* is able to enable data labeling by non-experts more accurately than the baseline approaches; further, an ensemble approach allows non-expert to contribute a significant portion of data to improve an existing AI model.

Acknowledgments

This work was funded by the Young Investigator Award by the Office of Naval Research (ONR N000142212188). We appreciate the anonymous reviewers for their comments on improving the manuscript, as well as the participants for their time and participation.

References

[1] Moloud Abdar, Farhad Pourpanah, Sadiq Hussain, Dana Rezazadegan, Li Liu, Mohammad Ghavamzadeh, Paul Fieguth, Xiaochun Cao, Abbas Khosravi, U Rajendra Acharya, et al. 2021. A review of uncertainty quantification in deep learning: Techniques, applications and challenges. *Information fusion* 76 (2021), 243–297.

[2] Sozan Mohammed Ahmed and Ramadhan J Mstafa. 2022. Identifying Severity Grading of Knee Osteoarthritis from X-ray Images Using an Efficient Mixture of Deep Learning and Machine Learning Models. *Diagnostics* 12, 12 (2022), 2939.

[3] Murat Seckin Ayhan and Philipp Berens. 2022. Test-time data augmentation for estimation of heteroscedastic aleatoric uncertainty in deep neural networks. In *Medical Imaging with Deep Learning*.

[4] Amir Banifatemi, Nicolas Miailhe, R Buse Çetin, Alexandre Cadain, Yolanda Lannquist, and Cyrus Hodes. 2021. Democratizing AI for humanity: A common goal. *Reflections on Artificial Intelligence for Humanity* (2021), 228–236.

[5] Katarzyna Borys, Yasmin Alyssa Schmitt, Meike Nauta, Christin Seifert, Nicole Krämer, Christoph M Friedrich, and Felix Nensa. 2023. Explainable AI in Medical Imaging: An overview for clinical practitioners–Saliency-based XAI approaches. *European journal of radiology* (2023), 110787.

[6] Zana Buçinca, Maja Barbara Malaya, and Krzysztof Z Gajos. 2021. To trust or to think: cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 1–21.

[7] Carrie J Cai, Emily Reif, Narayan Hegde, Jason Hipp, Been Kim, Daniel Smilkov, Martin Wattenberg, Fernanda Viegas, Greg S Corrado, Martin C Stumpe, et al. 2019. Human-centered tools for coping with imperfect algorithms during medical decision-making. In *Proceedings of the 2019 chi conference on human factors in computing systems*. 1–14.

[8] Chia-Ming Chang, Yi He, Xi Yang, Haoran Xie, and Takeo Igarashi. 2022. DualLabel: secondary labels for challenging image annotation. In *Graphics Interface 2022*.

[9] Chia-Ming Chang, Chia-Hsien Lee, and Takeo Igarashi. 2021. Spatial labeling: leveraging spatial layout for improving label quality in non-expert image annotation. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–12.

[10] Joseph Chee Chang, Saleema Amershi, and Ece Kamar. 2017. Revolt: Collaborative crowdsourcing for labeling machine learning datasets. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 2334–2346.

[11] Pingjun Chen. 2018. Knee osteoarthritis severity grading dataset. *Mendeley Data* 1 (2018), 21–23.

[12] Xianghui Chen, Shanjun Zhu, and Yindong Ji. 2000. Entropy based uncertainty measures for classification rules with inconsistency tolerance. In *Smc 2000 conference proceedings. 2000 ieee international conference on systems, man and cybernetics.'cybernetics evolving to systems, humans, organizations, and their complex interactions'(cat. no. 0*, Vol. 4. IEEE, 2816–2821.

[13] Victoria Clarke, Virginia Braun, and Nikki Hayfield. 2015. Thematic analysis. *Qualitative Psychology: A Practical Guide to Research Methods* (2015), 222–248.

[14] Andy Cockburn, Carl Gutwin, Joey Scarr, and Sylvain Malacria. 2014. Supporting novice to expert transitions in user interfaces. *ACM Computing Surveys (CSUR)* 47, 2 (2014), 1–36.

[15] Cathrine Damgaard, Trine Naja Eriksen, Dovile Juodelyte, Veronika Cheplygina, and Amelia Jiménez-Sánchez. 2023. Augmenting chest x-ray datasets with non-expert annotations. *arXiv preprint arXiv:2309.02244* (2023).

[16] Rocío Del Amor, Julio Silva-Rodríguez, and Valery Naranjo. 2023. Labeling confidence for uncertainty-aware histology image classification. *Computerized Medical Imaging and Graphics* 107 (2023), 102231.

[17] Michael Desmond, Michael Muller, Zahra Ashktorab, Casey Dugan, Evelyn Duesterwald, Kristina Brimijoin, Catherine Finegan-Dollak, Michelle Brachman, Aabhas Sharma, Narendra Nath Joshi, et al. 2021. Increasing the speed and accuracy of data labeling through an ai assisted interface. In *26th International Conference on Intelligent User Interfaces*. 392–401.

[18] Murat Dikmen and Catherine Burns. 2022. The effects of domain knowledge on trust in explainable AI and task performance: A case of peer-to-peer lending. *International Journal of Human-Computer Studies* 162 (2022), 102792.

[19] Yi-Li Fang, Hai-Long Sun, Peng-Peng Chen, and Ting Deng. 2017. Improving the quality of crowdsourced image labeling via label similarity. *Journal of Computer Science and Technology* 32 (2017), 877–889.

[20] Matthew Grissinger. 2019. Understanding human over-reliance on technology. *Pharmacy and Therapeutics* 44, 6 (2019), 320.

[21] Thomas Grote. 2021. Trustworthy medical AI systems need to know when they don't know. *Journal of medical ethics* 47, 5 (2021), 337–338.

[22] Andrzej Grzybowski, Kai Jin, and Hongkang Wu. 2024. Challenges of artificial intelligence in medicine and dermatology. *Clinics in dermatology* (2024).

[23] Hongyan Gu, Yuan Liang, Yifan Xu, Christopher Kazu Williams, Shino Magaki, Negar Khanlou, Harry Vinters, Zesheng Chen, Shuo Ni, Chunxu Yang, et al. 2023. Improving workflow integration with XPath: Design and evaluation of a human-AI diagnosis system in pathology. *ACM Transactions on Computer-Human Interaction* 30, 2 (2023), 1–37.

[24] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 770–778.

[25] Eyke Hüllermeier and Willem Waegeman. 2021. Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine Learning* 110 (2021), 457–506.

[26] Tim Hulsen. 2023. Explainable artificial intelligence (XAI): concepts and challenges in healthcare. *AI* 4, 3 (2023), 652–666.

[27] Vincent Humphrey, Matthew Rodell, and Annette Eicker. 2023. Using satellite-based terrestrial water storage data: A review. *Surveys in Geophysics* 44, 5 (2023), 1489–1517.

[28] Youngseung Jeon, Matthew K Hong, Yan-Ying Chen, Kalani Murakami, Jonathan Li, Xiang Anthony Chen, and Matthew Klenk. [n. d.]. Weaving ML with Human Aesthetic Assessments to Augment Design Space Exploration. ([n. d.]).

[29] Youngseung Jeon, Seung Gon Jeon, and Kyungsik Han. 2020. Better targeting of consumers: Modeling multifactorial gender and biological sex from Instagram posts. *User Modeling and User-Adapted Interaction* 30, 5 (2020), 833–866.

[30] Youngseung Jeon, Seungwan Jin, Patrick C Shih, and Kyungsik Han. 2021. FashionQ: an ai-driven creativity support tool for facilitating ideation in fashion design. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–18.

[31] Joanne M Jordan, G Fletcher Linder, Jordan B Renner, and John G Fryer. 1995. The impact of arthritis in rural populations. *Arthritis & Rheumatism: Official Journal of the American College of Rheumatology* 8, 4 (1995), 242–250.

[32] Daniel Kahneman. 2011. *Thinking, fast and slow.* macmillan.

[33] Daniel Kahneman, Shane Frederick, et al. 2002. Representativeness revisited: Attribute substitution in intuitive judgment. *Heuristics and biases: The psychology of intuitive judgment* 49, 49-81 (2002), 74.

[34] Mark D Kohn, Adam A Sassoon, and Navin D Fernando. 2016. Classifications in brief: Kellgren-Lawrence classification of osteoarthritis. *Clinical Orthopaedics and Related Research®* 474 (2016), 1886–1893.

[35] Ashnil Kumar, Shane Dyer, Jinman Kim, Changyang Li, Philip HW Leong, Michael Fulham, and Dagan Feng. 2016. Adapting content-based image retrieval techniques for the semantic annotation of medical images. *Computerized Medical Imaging and Graphics* 49 (2016), 37–45.

[36] Vivian Lai, Chacha Chen, Q Vera Liao, Alison Smith-Renner, and Chenhao Tan. 2021. Towards a science of human-ai decision making: a survey of empirical studies. *arXiv preprint arXiv:2112.11471* (2021).

[37] Jing Lin and Kee Yuan Ngiam. 2023. How data science and AI-based technologies impact genomics. *Singapore medical journal* 64, 1 (2023), 59–66.

[38] Matthew Lipman. 1987. Critical thinking: What can it be? *Analytic Teaching* 8, 1 (1987).

[39] Vivian Liu, Jo Vermeulen, George Fitzmaurice, and Justin Matejka. 2023. 3DALL-E: Integrating text-to-image AI in 3D design workflows. In *Proceedings of the 2023 ACM designing interactive systems conference*. 1955–1977.

[40] Mark Mazumder, Colby Banbury, Xiaozhe Yao, Bojan Karlaš, William Gaviria Rojas, Sudnya Diamos, Greg Diamos, Lynn He, Alicia Parrish, Hannah Rose Kirk, et al. 2022. Dataperf: Benchmarks for data-centric ai development. *arXiv preprint arXiv:2207.10062* (2022).

[41] Eduardo Mosqueira-Rey, Elena Hernández-Pereira, David Alonso-Ríos, José Bobes-Bascarán, and Ángel Fernández-Leal. 2023. Human-in-the-loop machine learning: a state of the art. *Artificial Intelligence Review* 56, 4 (2023), 3005–3054.

[42] Kumiyo Nakakoji, Yasuhiro Yamamoto, Shingo Takada, and Brent N Reeves. 2000. Two-dimensional spatial positioning as a means for reflection in design. In *Proceedings of the 3rd conference on Designing interactive systems: processes, practices, methods, and techniques*. 145–154.

[43] Naoki Otani, Yukino Baba, and Hisashi Kashima. 2015. Quality control for crowdsourced hierarchical classification. In *2015 IEEE International Conference on Data Mining*. IEEE, 937–942.

[44] Claudio Pacchiega. 2021. How Can Education Use Artificial Intelligence?: A Brief History of AI, Its Usages, Its Successes, and Its Problems When Applied to Education. In *Handbook of Research on Teaching With Virtual Environments and AI*. IGI Global, 558–590.

[45] Samir Passi and Mihaela Vorvoreanu. 2022. Overreliance on ai literature review. *Microsoft Research* (2022).

[46] Luise Pufahl and Mathias Weske. 2019. Batch activity: enhancing business process modeling and enactment with batch processing. *Computing* 101, 12 (2019), 1909–1933.

[47] Pranav Rajpurkar, Emma Chen, Oishi Banerjee, and Eric J Topol. 2022. AI in health and medicine. *Nature medicine* 28, 1 (2022), 31–38.

[48] George Robertson, Mary Czerwinski, Kevin Larson, Daniel C Robbins, David Thiel, and Maarten Van Dantzich. 1998. Data mountain: using spatial memory for document management. In *Proceedings of the 11th annual ACM symposium on User interface software and technology*. 153–162.

[49] Balaji Saibaba, Mandeep S Dhillon, Devendra K Chouhan, Rajendra K Kanojia, Mahesh Prakash, and Vikas Bachhal. 2015. Significant incidence of extra-articular tibia vara affects radiological outcome of total knee arthroplasty. *Knee Surgery & Related Research* 27, 3 (2015), 173.

[50] Ally Salim Jr, Megan Allen, Kelvin Mariki, Kevin James Masoy, and Jafary Liana. 2023. Understanding how the use of AI decision support tools affect critical thinking and over-reliance on technology by drug dispensers in Tanzania. *arXiv preprint arXiv:2302.09487* (2023).

[51] Max Schemmer, Patrick Hemmer, Niklas Kühl, Carina Benz, and Gerhard Satzger. 2022. Should I follow AI-based advice? Measuring appropriate reliance in human-AI decision-making. *arXiv preprint arXiv:2204.06916* (2022).

[52] Max Schemmer, Niklas Kuehl, Carina Benz, Andrea Bartos, and Gerhard Satzger. 2023. Appropriate reliance on AI advice: Conceptualization and the effect of explanations. In *Proceedings of the 28th International Conference on Intelligent User Interfaces*. 410–422.

[53] Sascha Seifert, Michael Kelm, Manuel Moeller, Saikat Mukherjee, Alexander Cavallaro, Martin Huber, and Dorin Comaniciu. 2010. Semantic annotation of medical images. In *Medical Imaging 2010: Advanced PACS-based Imaging Informatics and Therapeutic Applications*, Vol. 7628. SPIE, 43–50.

[54] Ramprasaath R Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. 2017. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision*. 618–626.

[55] Burr Settles. 2009. Active learning literature survey. (2009).

[56] Monideepa Tarafdar, Cynthia M Beath, and Jeanne W Ross. 2019. Using AI to enhance business operations. *MIT Sloan Management Review* 60, 4 (2019), 37–44.

[57] John W Tukey. 1949. Comparing individual means in the analysis of variance. *Biometrics* (1949), 99–114.

[58] Bas HM Van der Velden, Hugo J Kuijf, Kenneth GA Gilhuijs, and Max A Viergever. 2022. Explainable artificial intelligence (XAI) in deep learning-based medical image analysis. *Medical Image Analysis* 79 (2022), 102470.

[59] Xinru Wang and Ming Yin. 2021. Are explanations helpful? a comparative study of the effects of explanations in ai-assisted decision-making. In *Proceedings of the 26th International Conference on Intelligent User Interfaces*. 318–328.

[60] Maximiliane Wilkesmann and Uwe Wilkesmann. 2011. Knowledge transfer as interaction between experts and novices supported by technology. *Vine* 41, 2 (2011), 96–112.

[61] Michael Yeomans, Anuj Shah, Sendhil Mullainathan, and Jon Kleinberg. 2019. Making sense of recommendations. *Journal of Behavioral Decision Making* 32, 4 (2019), 403–414.

[62] Tianwei Zhang, Lequan Yu, Na Hu, Su Lv, and Shi Gu. 2020. Robust medical image segmentation from non-expert annotations with tri-network. In *Medical Image Computing and Computer Assisted Intervention–MICCAI 2020: 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings, Part IV 23*. Springer, 249–258.

A Appendix

A.1 Table and figure

Table 4: Demographic information of the pilot study.

Number	Major	Age	Gender
P_p^1	Philosophy (BA)	22	Male
P_p^2	Computer Science (MS)	22	Male
P_p^3	Computer Science (BS)	22	Male
P_p^4	Illustration (BA)	23	Female
P_p^5	Computer Science (MS)	22	Male
P_p^6	Data Science (BS)	20	Male

Table 5: Demographic information of the participants in user studies and interviews (total: 28 participants for first and 12 participants for second). Education is based on enrollment status.

Demographic group		First		Second	
		N	%	N	%
Gender	Female	10	36%	1	8%
	Male	18	64%	11	92%
Age	18-25	14	50%	12	100%
	26-35	13	46%	0	0%
	36-45	1	4%	0	0%
Education	Doctoral degree	11	39%	1	8%
	Master's degree	8	29%	4	33%
	Undergraduate	9	32%	7	59%
Major	Engineering	14	50%	9	75%
	Science	7	25%	0	0%
	Social Sciences	4	14%	1	8%
	Interdisciplinary	3	11%	2	17%

B Uncertainty and accuracy

This section shows 1) how to define uncertainty groups and 2) their accuracy results in the first user study. We built uncertainty groups and evaluated their accuracy to identify a possible direction in which non-experts who lack specialized knowledge can contribute to the labeling process.

B.1 Uncertainty in the first user study

There are two types of uncertainty estimation [25]: 1) Epistemic uncertainty and 2) Aleatoric uncertainty. Considering that our goal is to calculate the uncertainty of the dataset, we chose the Aleatoric uncertainty, which is the variability inherent in the observed data, not in AI models. To implement the setting for Aleatoric uncertainty, we referred to previous work [3] where the authors examine the degree of change in model predictions when modifications such as clipping, flipping horizontally and vertically, and color transformation. We introduced randomness to the data variation and obtained

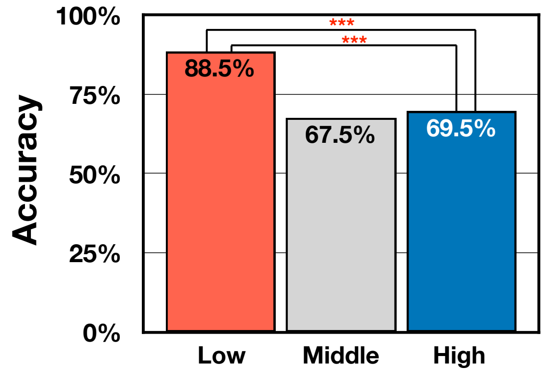

Figure 9: Bar plots showing the accuracy differences between the uncertainty groups (Low, Middle, and High) in experimental and control groups ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$).

the consistency level through 100 predictions. We calculate the entropy value defined as follows [12].

$$H(p) = -[p \log(p) + (1 - p) \log(1 - p)] \tag{1}$$

where H means the result of entropy and p is the mean of the probability of belonging to a class when making 100 predictions. When p is 0.5, entropy reaches its maximum value of 1. Using the model (Sec. 3.2), we extracted the uncertainty score of (a) the Total dataset (50 images), sorted the uncertainty score by descending order, and then divided it into three groups: The Low-uncertainty group (17 images, mean=0.635, SD=0.053), the Middle-uncertainty group (16 images, mean=0.686, SD=0.002), and the High-uncertainty group (17 images, mean=0.692, SD=0.001). Figure 10 shows the representative examples of three uncertainty groups: Low, Middle, and High.

To demonstrate the relationship between uncertainty and accuracy of non-experts' labeling, we evaluate whether there are significant differences in labeling accuracy between the uncertainty groups (Low, Middle, and High). A one-way ANOVA was conducted to measure and post-hoc comparisons were also conducted using Tukey's correction [57]. Figure 9 shows that there is a significant difference in accuracy between the uncertainty groups (Low, Middle, and High) ($F(2,25)$=33.7, p<.001). According to the post-hoc analysis, there are significant differences between the Low and remaining two groups: 1) the Middle: p<.001 and 2) the High: p<.001. This might mean that for non-experts, the ability to label the low-uncertainty group more accurately than the middle and high-uncertainty groups signifies.

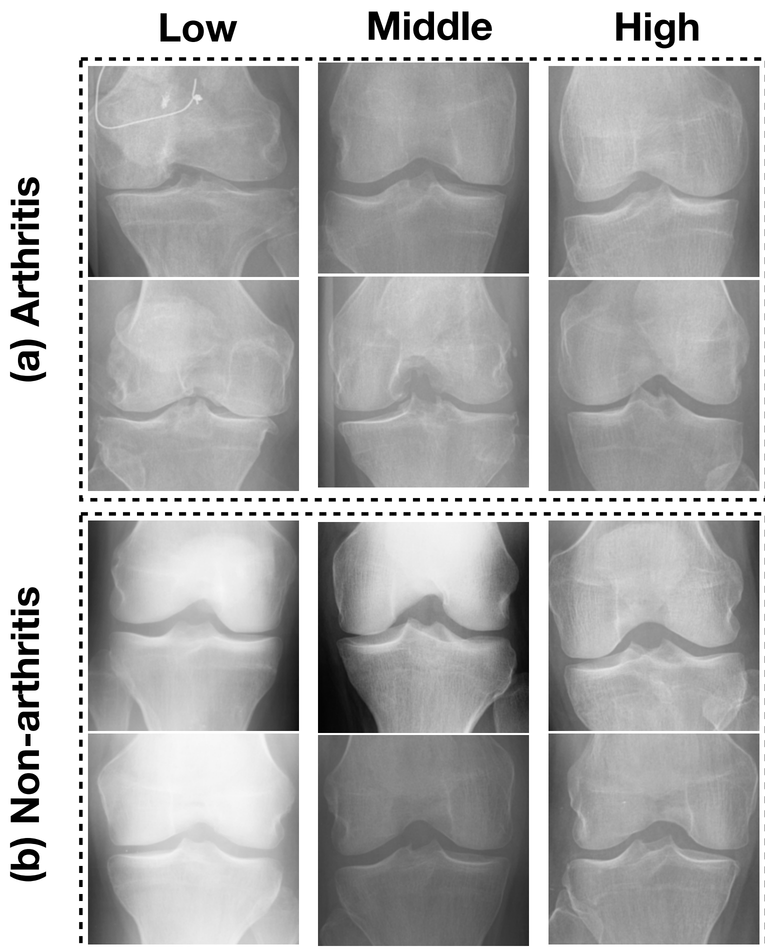

Figure 10: The representative examples of three uncertainty groups (Low, Middle, and High) in (a) Arthritis and (b) Non-arthritis. In the case of (a) Arthritis, the images of the low uncertainty groups clearly appear to possess all four features of arthritis. On the other hand, The images of the Middle- and High-uncertainty groups appear to have one or two of those features. In the case of (b) Non-arthritis, the images of the low uncertainty groups clearly appear to not possess that feature. On the other hand, The images of the Middle- and High-uncertainty groups appear to have one or two of those features.